

SELF-DRIVING CAR

INTELLIGENT AGENT

Student Information

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Section: B

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SELF-DRIVING CAR

As an Intelligent Agent in Artificial Intelligence

ENVIRONMENT Road • Traffic • Pedestrians • Signals	INTELLIGENT AGENT AI Perception + Planning + Decision Making	ACTIONS Steering • Braking • Acceleration
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Prepared for: Artificial Intelligence / Intelligent Agents

Topic: Self-Driving Car Intelligent Agent

1. Introduction

A self-driving car is a real-world example of an **intelligent agent**. An intelligent agent is a system that perceives its environment through sensors, processes information, makes decisions, and performs actions through actuators to achieve a goal. A self-driving car continuously observes the road and traffic, predicts what may happen next, and chooses safe driving actions.

2. What is an Intelligent Agent?

An intelligent agent is an entity that **perceives** its environment using sensors and **acts** on that environment using actuators. The agent tries to select actions that maximize its performance according to a defined performance measure.

Basic Agent Model

Percepts	Agent	Actions
Camera, radar, lidar, GPS, speed sensors	AI system / decision-making software	Steering, brake, accelerator, indicators

3. Self-Driving Car as an Intelligent Agent

The self-driving car behaves like an agent because it receives information from its surroundings, interprets that information, decides what should happen next, and controls the vehicle. For example, when a pedestrian is detected crossing the road, the system can identify the pedestrian, estimate the risk, plan a response, and apply the brakes if necessary.

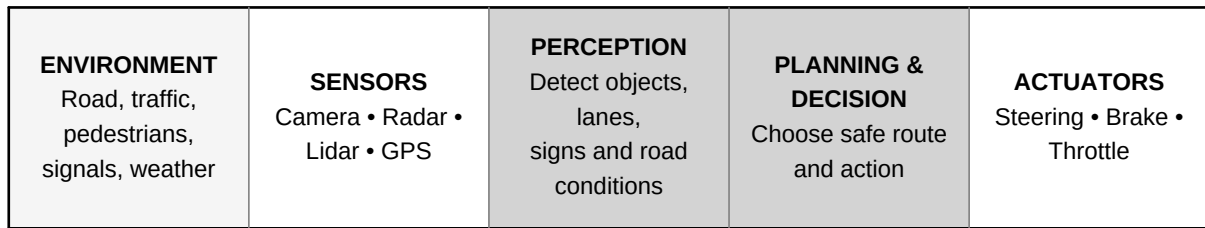
4. PEAS Description

PEAS stands for **Performance measure, Environment, Actuators, and Sensors**. It is commonly used to specify an intelligent agent's task environment.

PEAS Element	Self-Driving Car
Performance Measure	Safety, collision avoidance, obeying traffic rules, reaching destination, passenger comfort, fuel/energy efficiency
Environment	Roads, highways, intersections, traffic lights, vehicles, pedestrians, cyclists, weather, road signs and obstacles
Actuators	Steering system, accelerator/throttle, brakes, gear control, turn signals and other vehicle controls.
Sensors	Cameras, radar, lidar, GPS, ultrasonic sensors, wheel-speed sensors, inertial measurement sensors and other sensors

5. Graphical Representation of the Agent

The following block diagram shows the complete perception–decision–action cycle of a self-driving car.



Feedback loop: the car acts on the environment, observes the new situation, and repeats the cycle continuously.

6. Working Process

- 1. Sensing:** Sensors collect information about the car and its surroundings.
- 2. Perception:** AI models identify lanes, vehicles, pedestrians, signs, traffic lights and obstacles.
- 3. Localization:** GPS and other sensors help estimate the car's position and movement.
- 4. Prediction:** The system estimates how nearby vehicles and pedestrians may move.
- 5. Planning:** A safe route and short-term driving trajectory are selected.
- 6. Decision Making:** The agent decides actions such as slowing, stopping, turning or changing lanes.
- 7. Control:** Vehicle actuators execute the selected actions.
- 8. Feedback:** New sensor data is used to evaluate the result and update the next decision.

7. Types of Intelligent Behavior

A self-driving car can be viewed primarily as a **model-based, goal-based and utility-based agent**. It maintains an internal representation of the world, has goals such as reaching a destination safely, and must balance competing objectives such as safety, comfort, travel time and energy efficiency. Learning methods may also improve perception or prediction, but safe operation still depends on carefully designed rules, models and controls.

8. Real-World Example

Scenario: The car is travelling on a city road and a pedestrian suddenly approaches a crossing.

Stage	Agent Response
Sense	Camera/radar detect the pedestrian and surrounding traffic.
Understand	Perception software identifies the pedestrian and estimates location and movement.
Predict	The system estimates whether the pedestrian may enter the vehicle's path.
Plan	The planner selects a safe response, such as reducing speed or stopping.
Act	The control system commands braking and maintains steering.
Verify	Sensors confirm the vehicle has slowed/stopped and the situation is safe before continuing.

9. Advantages

- Potential to reduce crashes caused by human distraction, fatigue or impaired attention.
- Continuous sensing and rapid computation can support timely responses.
- Can assist people who cannot drive and improve mobility.
- May optimize routes, traffic flow and energy use.
- Can provide consistent execution of programmed driving policies.

10. Challenges and Limitations

- Unusual or unexpected road situations can be difficult to handle.
- Heavy rain, fog, snow, glare or sensor obstruction can reduce perception quality.
- Complex traffic behavior requires accurate prediction and robust decision-making.
- Safety, cybersecurity, privacy, testing and regulatory requirements are significant.
- High-quality sensors, computing hardware, software validation and maintenance can be expensive.

11. Conclusion

The self-driving car is a strong example of an intelligent agent because it follows the complete agent cycle: **perceive** → **understand** → **decide** → **act** → **receive feedback**. Its sensors provide percepts, AI software makes decisions, and actuators change the vehicle's behavior. Using the PEAS framework and the graphical model makes it easy to understand how an intelligent agent operates in a complex real-world environment.

12. Key Terms

Intelligent Agent	A system that perceives an environment and acts to achieve goals.
Sensor	A component that gathers information from the environment or vehicle.
Actuator	A mechanism that performs an action in the physical world.
PEAS	Performance measure, Environment, Actuators and Sensors.
Perception	Interpreting sensor data to understand the surroundings.

Planning	Selecting a future path or sequence of actions.
Feedback	Using new observations to update the next action.

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